

Software Requirement Specification

NutriSage: Smart Nutrition Assistant for Food Analysis and Personalized Meal Guidance



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Abstract

Nutrition data is often available in raw numbers (calories, proteins, fats, etc.), but most people struggle to interpret what those numbers mean for their health. This project aims to develop a Smart Nutrition Assistant, **NutriSage**, that makes nutrition data more understandable, personalized, and actionable.

We will use the USDA Foundation Foods dataset as the backbone of a system that explains food nutrients in a meaningful health context (e.g., “This apple provides 5% of daily energy needs and is heart-healthy due to its potassium content”). The system will accept flexible, multi-modal inputs, allowing users to type meals in natural language like “2 cups oatmeal + 1 banana,” which is then converted into detailed nutrition insights.

A core feature is a chatbot assistant, powered by Retrieval-Augmented Generation (RAG), that answers food-related questions and suggests healthier alternatives based on a trusted knowledge base. The platform will also provide essential tools for meal planning, food comparison, and daily nutrition tracking, complete with smart recommendations to help users balance their nutrient intake.

By combining structured food data with AI-driven interpretability, NutriSage aims to make expert nutrition advice more accessible, personalized, and meaningful for everyone.

Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
EDA	Exploratory Data Analysis
LLM	Large Language Model
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
RDA	Recommended Daily Allowance
SRS	Software Requirements Specification
TLS	Transport Layer Security
UI	User Interface
USDA	United States Department of Agriculture
WCAG	Web Content Accessibility Guidelines

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1 Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a comprehensive description of the **NutriSage** platform—an intelligent nutrition assistant designed to help users understand their food better, track daily intake, and receive personalized dietary suggestions. In an era of nutritional misinformation, this system aims to be a source of truth, empowering users to make informed decisions.

While nutrition data is widely available, it often exists as raw numbers (calories, protein, fats) that are difficult for the average person to interpret in a meaningful health context. NutriSage aims to bridge this gap by combining the authoritative USDA Foundation Foods dataset with AI-driven interpretation, making nutritional insights understandable, personalized, and actionable. The primary functions of key macro and micronutrients are summarized in Table 1 and Table 2, respectively.

Table 1: Key Macronutrients and Their Primary Functions

Macronutrient	Primary Function
Protein	Essential for building and repairing tissues, making enzymes and hormones. It is a critical building block of bones, muscles, cartilage, skin, and blood.
Carbohydrates	The body’s main source of energy. They are broken down into glucose, which fuels the brain, kidneys, heart muscles, and central nervous system.
Fats	A source of essential fatty acids, which the body cannot make itself. Fats help the body absorb vitamins A, D, and E, and are vital for energy storage and insulation.

Table 2: Key Micronutrients and Their Roles

Micronutrient	Primary Role in the Body
Vitamin C	Acts as an antioxidant, necessary for the growth, development, and repair of all body tissues. It’s involved in many body functions, including the formation of collagen and absorption of iron.
Iron	A vital component of hemoglobin, the substance in red blood cells that carries oxygen from your lungs to transport it throughout your body. A lack of iron can lead to anemia.
Calcium	Crucial for building and maintaining strong bones. The heart, muscles, and nerves also need calcium to function properly.

1.2 Scope of the Project

NutriSage is a comprehensive web and mobile platform that provides a suite of tools to demystify nutrition. The project’s scope is defined by its core features, target audience, and intended

impact.

1.2.1 Core Features

- **Interpretability Layer:** This core component translates complex numerical data into simple, actionable advice. It goes beyond just showing numbers by contextualizing them (e.g., "This amount of potassium supports heart health") and calculates a proprietary "nutrient diversity score" to encourage balanced eating.
- **Multi-Modal Input:** To ensure ease of use, the system accepts natural language queries for meal logging (e.g., "ate a salad with chicken and vinaigrette"). Future development will incorporate image recognition, allowing users to simply take a photo of their meal for automatic analysis.
- **AI-Powered Chatbot:** A conversational assistant built with Retrieval-Augmented Generation (RAG) acts as a personal nutritionist. It answers user questions by drawing from a curated knowledge base of trusted nutritional science, preventing the spread of misinformation and providing evidence-based suggestions.
- **Daily Nutrition Tracker:** A user-friendly dashboard provides a visual summary of daily intake, tracking calories, macros, and key micronutrients against scientifically established recommended daily allowances (RDA).
- **Supporting Tools (Phase 2):** To further enhance user capabilities, the platform will later include a Meal/Diet Planner for creating structured eating schedules and a Food Comparison Tool for making informed choices between different food items.

1.2.2 Target Users

- Health-conscious individuals seeking to understand and scientifically optimize their diet.
- Users with specific dietary goals, such as weight management, muscle gain, or addressing nutrient deficiencies, who require tailored guidance.
- Individuals looking for a simple, low-effort tool to track their daily food consumption and improve their eating habits.

1.2.3 Expected Outcomes

- **Improved nutritional literacy:** Users will gain a deeper understanding of the food they eat and its impact on their health.
- **Simplified dietary management:** The platform will make meal tracking and analysis an effortless daily habit.
- **Better health outcomes:** By providing personalized, actionable recommendations, the system aims to help users achieve their health goals.
- **Accessible science:** The platform will make complex nutritional science accessible and practical for a general audience.

1.3 Literature Review

The development of NutriSage is informed by established research in nutritional science, data engineering, and artificial intelligence.

Nutritional Data Accessibility: The USDA FoodData Central, particularly the Foundation Foods dataset, provides highly detailed and verified nutrient profiles for a wide range of foods. However, studies show a significant gap between data availability and public comprehension. Most individuals struggle to translate quantitative data (e.g., "25g of protein") into qualitative health impact [1]. Our project directly addresses this "interpretability gap" by building a semantic layer over this raw data.

Natural Language Processing (NLP) in Healthcare: Recent advancements in NLP, particularly with transformer-based models like BERT, have enabled systems to understand complex, unstructured text. Research by Liang et al. [4] and Collins et al. [2] demonstrates the effectiveness of NLP in extracting food and health-related entities from unstructured text, such as dietetic records or user logs. NutriSage will leverage these techniques to accurately parse user meal descriptions like "a bowl of oatmeal with a banana and walnuts," identifying both the foods and their quantities.

Computer Vision for Food Recognition: As a future extension, image-based logging presents a significant usability improvement. The "Im2Calories" project [5] demonstrated the feasibility of estimating calories from food images, highlighting the potential of deep learning models in this domain. NutriSage plans to build upon this research to classify food items and estimate portion sizes from user-submitted photos.

Retrieval-Augmented Generation (RAG) for Factual AI: Standard Large Language Models (LLMs) can sometimes produce inaccurate or unverified information ("hallucinate"). The RAG framework mitigates this by grounding the model's responses in a specific, trusted knowledge base [3]. For NutriSage, this is a critical feature, ensuring the chatbot's answers to nutrition questions are based on established scientific literature and dietary guidelines, not just probabilistic text generation.

Personalized Nutrition: The concept that different individuals respond differently to the same foods is a growing area of research. Work by Zeevi et al. [6] showed that blood glucose responses to meals are highly individual. While deep personalization is a long-term goal, NutriSage's design lays the groundwork by enabling detailed tracking that could eventually fuel more personalized recommendation algorithms.

Identified Gaps:

- Existing applications often focus on simplistic calorie counting but lack a deep, educational interpretability layer that explains the *why* behind the numbers.
- Few systems effectively integrate a trusted, RAG-based conversational agent for personalized, safe, and scientifically-backed advice.
- The concept of a "nutrient diversity score" to holistically evaluate and promote balanced eating is a novel feature not widely implemented in consumer applications.

2 System Requirements

2.1 Functional Requirements

This section outlines the core functions of the NutriSage system, grouped by feature. Table 3 provides a high-level summary.

Table 3: Summary of Functional Requirements

ID	Feature	Description
FR1	Interpretability Layer	Translates raw nutrient data into simple explanations, health risks, and a nutrient diversity score.
FR2	Multi-Modal Input	Allows users to log meals via natural language text, with future scope for image recognition.
FR3	Chatbot with RAG	Provides a conversational AI for nutrition Q&A, explanations, and healthy alternative suggestions.
FR4	Daily Nutrition Tracker	Enables users to log meals and view a dashboard of daily calories, macros, and micronutrient intake.
FR5	Meal/Diet Planner	(Phase 2) Generates personalized meal plans based on user goals and preferences.
FR6	Food Comparison Tool	(Phase 2) Offers a side-by-side comparison of foods with contextual health information.

2.1.1 Interpretability Layer

- FR1.1** The system shall display a detailed breakdown of macronutrients (protein, fats, carbohydrates) and key micronutrients (e.g., Vitamin C, Iron, Potassium) for any given food or meal. The display will include absolute values (grams, mg) and the percentage of the Recommended Daily Allowance (%RDA).
- FR1.2** For each major nutrient, the system shall provide simple, plain-language explanations of its primary function in the body (e.g., "Potassium is an electrolyte that helps regulate blood pressure and muscle contractions"). This content must be easily accessible, perhaps via a tooltip or a collapsible section.
- FR1.3** The system shall use a color-coded system (e.g., red, amber, green) to highlight nutrients that are significantly high or low relative to dietary guidelines (e.g., high sodium, low fiber), providing a quick visual assessment of the food's health profile.
- FR1.4** The system shall calculate and display a proprietary "nutrient diversity score." This score will be computed based on the breadth of essential micronutrients present in a meal, encouraging users to consume a wider variety of foods rather than focusing solely on calories or macros.

2.1.2 Multi-Modal Input System

- FR2.1** The system must parse complex, multi-item meal descriptions entered in natural language (e.g., "For breakfast I had 2 scrambled eggs with a slice of whole wheat toast and a glass of orange juice").
- FR2.2** The NLP model shall be trained to recognize food entities, quantities (e.g., "2", "a slice", "a glass"), units (e.g., "cups", "grams"), and preparation methods (e.g., "scrambled", "grilled") to accurately map the input to the food database.
- FR2.3** (Future Scope) The system shall provide a clean UI for users to upload a photo of their meal. The system must support common image formats (JPEG, PNG).

FR2.4 (Future Scope) The system's computer vision model must be able to identify multiple food items within a single, complex image (e.g., a plate with chicken, rice, and broccoli) and estimate their volume to approximate portion sizes.

2.1.3 Chatbot with RAG

FR3.1 The system shall provide a conversational chatbot capable of understanding and responding to a wide range of nutrition-related questions, from simple facts ("How much Vitamin C is in an orange?") to complex queries ("What are good protein sources for a vegetarian diet?").

FR3.2 The chatbot's knowledge base for the RAG model shall be built exclusively from trusted sources, including government health publications, peer-reviewed scientific journals, and established nutritional textbooks, to ensure the information provided is accurate and safe.

FR3.3 The chatbot must maintain conversational context, allowing users to ask follow-up questions without having to restate the initial query (e.g., User: "Is salmon healthy?" Bot: "Yes, it's high in Omega-3s." User: "What do those do?").

FR3.4 The chatbot's suggestions for healthier alternatives shall be context-aware, considering the original food item's role in a meal (e.g., suggesting baked sweet potato fries instead of deep-fried potato fries as a side dish).

2.1.4 Daily Nutrition Tracker

FR4.1 Users shall be able to log breakfast, lunch, dinner, and snacks for any given day. The system must also allow for editing and deleting past entries to correct mistakes.

FR4.2 The tracking dashboard shall feature clear visualizations, including a pie chart for macronutrient distribution (protein, carbs, fat), and a series of progress bars showing the percentage of RDA achieved for at least 10 key micronutrients.

FR4.3 The system shall generate daily summaries and intelligent recommendations. These recommendations will be specific and actionable, such as suggesting a Vitamin C-rich food if the user's intake is low, or advising a lower-sodium option for dinner if their daily sodium limit is being approached.

2.1.5 Meal/Diet Planner (Phase 2)

FR5.1 The system shall generate personalized meal plans based on a detailed user profile, including age, weight, height, activity level, dietary goals (weight loss, muscle gain), and dietary restrictions (e.g., vegetarian, gluten-free).

FR5.2 The meal planner's algorithm shall prioritize the nutrient diversity score to ensure the generated plans are balanced and nutritionally complete, preventing deficiencies that can arise from restrictive diets.

2.1.6 Food Comparison Tool (Phase 2)

- FR6.1** The system shall allow users to select two or more food items from the database and view their nutritional information in a side-by-side tabular format for easy comparison.
- FR6.2** The comparison view shall automatically highlight significant differences, such as one item being much higher in sugar while another is higher in fiber, and provide a brief text summary explaining the health implications of these differences.

2.2 Non-Functional Requirements

This section details the quality attributes, design constraints, and standards the NutriSage system must adhere to, as summarized in Table 4.

Table 4: Summary of Non-Functional Requirements

ID	Category	Description
NFR1	Performance	Defines system responsiveness, throughput, and speed under various load conditions.
NFR2	Usability	Focuses on the ease of use, user interface design, and overall user experience.
NFR3	Reliability	Specifies system availability, data integrity, and error handling capabilities.
NFR4	Security	Outlines requirements for data protection, user privacy, and secure authentication.

2.2.1 Performance Requirements

- NFR1.1** All API calls for nutrient data retrieval for a single food item shall have a 95th percentile response time of less than 500ms.
- NFR1.2** Natural language meal parsing for an average sentence length (10-15 words) shall complete within 3 seconds under normal server load.
- NFR1.3** The chatbot's time-to-first-response for a new query shall be under 3 seconds to ensure a fluid, conversational feel. The system must support at least 100 concurrent chat sessions without significant degradation in performance.

2.2.2 Usability Requirements

- NFR2.1** The user interface shall be intuitive, clean, and require minimal training for a non-technical user. A new user should be able to successfully log their first meal within 60 seconds of signing up.
- NFR2.2** The application must be fully responsive, providing an optimized and consistent user experience on major web browsers (Chrome, Firefox, Safari) and on mobile devices (iOS and Android).
- NFR2.3** The system shall adhere to Web Content Accessibility Guidelines (WCAG) 2.1 Level AA to ensure it is usable by people with disabilities, including support for screen readers and keyboard navigation.

2.2.3 Reliability Requirements

- NFR3.1** The system's core services (meal logging, data lookup) shall maintain a monthly uptime of 99.5% or higher.
- NFR3.2** The system must implement robust error handling. For instance, if the NLP model cannot parse a food item, it should provide the user with suggestions or prompt for clarification rather than returning a cryptic error.
- NFR3.3** All user data must be backed up daily to a separate geographical location to prevent data loss in case of a server failure or disaster. The recovery time objective (RTO) shall be less than 24 hours.

2.2.4 Security Requirements

- NFR4.1** All sensitive user data, including personal information, health goals, and dietary logs, shall be encrypted both in transit (using TLS 1.2 or higher) and at rest (using AES-256 encryption).
- NFR4.2** The system must implement secure user authentication, including password complexity requirements and protection against brute-force attacks.
- NFR4.3** User data shall not be shared with any third parties without explicit, opt-in consent from the user for each specific instance of data sharing, in accordance with privacy best practices.

2.3 AI-Specific Requirements

This section outlines requirements unique to the Artificial Intelligence components of NutriSage, summarized in Table 5.

Table 5: Summary of AI-Specific Requirements

ID	Category	Description
AIR1	Model Performance	Defines the quantitative success criteria for the AI models (NLP, RAG, CV).
AIR2	Ethical & Safety	Establishes the safety guardrails and ethical principles for providing AI-driven health advice.

2.3.1 Model Performance

- AIR1.1** The Natural Language Processing (NLP) model for meal parsing must achieve at least 90% accuracy in identifying food entities and quantities from user input on a designated test dataset.
- AIR1.2** The Retrieval-Augmented Generation (RAG) chatbot must provide citations from its knowledge base for at least 95% of factual claims to ensure transparency and traceability.
- AIR1.3** The RAG chatbot must have a mechanism to detect and gracefully handle queries that are outside its knowledge domain, responding with a message like "I can only answer questions about nutrition."

AIR1.4 (Future Scope) The computer vision model for food recognition shall achieve a top-3 accuracy of 85% in identifying common food items from user-submitted images.

2.3.2 Ethical and Safety Requirements

AIR2.1 The system shall include a clear and prominent disclaimer stating that it is a nutritional guide and not a substitute for professional medical advice, especially for users with medical conditions.

AIR2.2 The chatbot shall be programmed to refuse to provide guidance on medical diagnoses, eating disorders, or extreme diets. In such cases, it will recommend consulting a registered dietitian or doctor.

AIR2.3 All personalized recommendations generated by the system shall be based on established public health guidelines (e.g., RDA values) and avoid promoting fad diets or unverified nutritional claims.

3 Data Preprocessing

The data preprocessing phase is critical for transforming the raw MM-Food-100K.csv dataset into a clean, reliable, and efficient format suitable for model training and application use. The pipeline, as detailed in the project notebook, is designed to be robust, efficient, and resumable.

3.1 Initial Setup and Configuration

This preliminary step involves defining the environment in which the entire preprocessing pipeline will run. Key activities include:

- **Path Definition:** Setting up file paths for the raw input CSV, the main output directory, and subdirectories for storing processed images.
- **Parameter Configuration:** Defining crucial parameters such as the number of concurrent asynchronous requests for image downloading, maximum image dimensions for resizing, JPEG quality for compression, and the minimum image resolution to filter out low-quality thumbnails.
- **Schema Definition:** Specifying the final set of columns and nutrient keys that will constitute the clean dataset.

3.2 Feature Engineering and Text Cleaning

The raw dataset contains semi-structured and unstructured data in several columns. This stage focuses on parsing this data into usable features with a series of defined helper functions.

- **Nutrient Parsing:** The `nutritional_profile` column, which contains JSON data as a string, is safely parsed using a helper function that handles malformed entries. Key nutrient values are then extracted into new, strongly-typed columns in the dataframe.

- **Text Normalization:** Textual data from columns like `dish_name`, `ingredients`, and `cooking_method` undergoes a standardization process. This involves converting all text to lowercase and removing any characters that are not alphanumeric, hyphens, or apostrophes, ensuring consistency for analysis and modeling.
- **Ingredient Standardization:** The `ingredients` column, often a string representation of a list, is parsed using `ast.literal_eval`. Each ingredient is then individually cleaned, and the final list is sorted and deduplicated to create a canonical representation for each dish.
- **Derived Features:** New, insightful features are created from existing data. Specifically, the `portion_size` string is parsed using regular expressions to extract a numerical `total_grams` value. This new feature is then used to calculate nutrient density metrics (e.g., `calories_kcal_per_100g`), allowing for fair comparisons between dishes of different sizes.

3.3 Dataset Sampling

To make the computationally expensive image processing step more manageable, the full dataset is down-sampled while carefully maintaining its statistical properties.

- **Stratified Sampling Methodology:** Rather than taking a simple random sample, we employ stratified sampling using the `train_test_split` function from the `scikit-learn` library. The dataset is partitioned based on the `food_type` column, and a proportional sample is taken from each category. This guarantees that the resulting 60,000-record subset has the same distribution of food types as the original 100,000-record dataset, making it a smaller yet highly representative sample for further processing. This is demonstrated in Table 6.

Table 6: Stratified Sampling Distribution Preservation

Food Type	Original Distribution (%)	Sample Distribution (%)
Homemade Food	46.56	46.56
Restaurant Food	35.46	35.46
Raw Vegetables & Fruits	9.36	9.36
Packaged Food	8.35	8.35
Others	0.27	0.27

3.4 Image Processing Pipeline

This is the most computationally intensive step, involving the download and processing of tens of thousands of images. The outcomes of this pipeline are summarized in Table 7.

- **Asynchronous Downloading:** An asynchronous, concurrent approach is used to download images from their URLs, significantly speeding up the process compared to a sequential method.
- **Image Standardization:** Each downloaded image is converted to RGB format, resized to a maximum dimension of 512 pixels, and compressed as a high-quality JPEG. This ensures uniformity and reduces storage requirements.

- **Duplicate Detection:** A perceptual hash (phash) is generated for each valid image. This allows for the identification and flagging of visually identical or near-identical images, which is crucial for creating a diverse training set.
- **Resumability and Checkpointing:** The process is designed to be resumable. It operates in batches and saves its progress to a CSV file after each batch. If the script is interrupted, it can be restarted and will only attempt to process the images that were not completed in the previous run.

Table 7: Image Processing Status Summary

Status	Count
Download Error	47,534
OK (Processed)	10,996
Not an Image	983
Too Small	486
Processing Error	1

3.4.1 Batch Processing Performance

The image download and processing task was executed in 12 batches of 5,000 images each. As shown in Table 8, the time taken for each batch varied significantly. The initial batches took longer, while the final batches were extremely fast. This is a direct result of the resumability feature of the script; the later runs found that most images had already been processed and cached from previous runs, so they completed very quickly by skipping the download step for existing files.

Table 8: Image Processing Batch Performance

Batch Number	Images in Batch	Time Taken
1	5,000	18m 14s
2	5,000	22m 17s
3	5,000	20m 41s
4	5,000	11m 10s
5	5,000	18m 44s
6	5,000	23m 44s
7	5,000	21m 20s
8	5,000	8m 28s
9	5,000	14s
10	5,000	14s
11	5,000	9s
12	5,000	2s

3.5 Final Dataset Assembly

The final step involves collating all processed information into a final, clean dataset.

- **Filtering:** The main dataframe is filtered to include only the records for which an image was successfully downloaded and processed. Rows with invalid URLs, download errors, or images that were too small are discarded.

- **Schema Finalization:** The dataframe is pruned to keep only the clean, final columns. Columns are renamed for clarity and consistency.
- **Output Generation:** The resulting clean dataframe is saved as `MM-Food-Cleaned-Final.csv`, which serves as the definitive data source for all subsequent EDA and model training tasks.

4 Exploratory Data Analysis (EDA)

Following the preprocessing phase, Exploratory Data Analysis (EDA) is conducted on the clean `MM-Food-Cleaned-Final.csv` dataset. The goal of EDA is to uncover patterns, identify anomalies, and gain a deep understanding of the data's structure and content before proceeding to model development.

4.1 Categorical Feature Analysis

The analysis begins by examining the distribution of the primary categorical features. Figure 1 shows the count of dishes by food type. The dataset is primarily composed of 'homemade food' and 'restaurant food', which together account for the vast majority of entries. This indicates a good balance between professionally prepared and home-cooked meals.

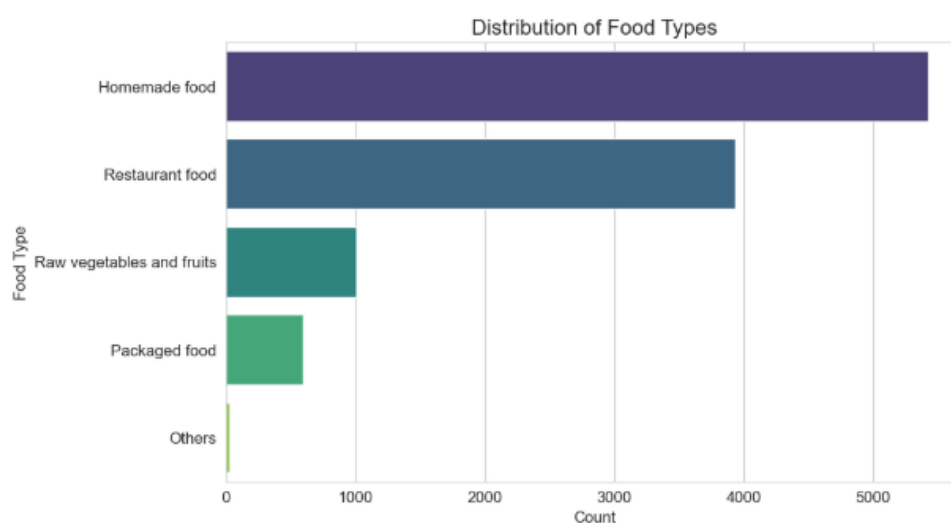


Figure 1: Distribution of Food Types

Figure 2 illustrates the top 10 most common cooking methods. Methods like 'stir-frying', 'boiling', and 'frying' are highly represented, which provides insight into the culinary nature of the dishes in the dataset. The presence of 'raw' aligns with the 'raw vegetables and fruits' food type and highlights a segment of unprocessed foods.

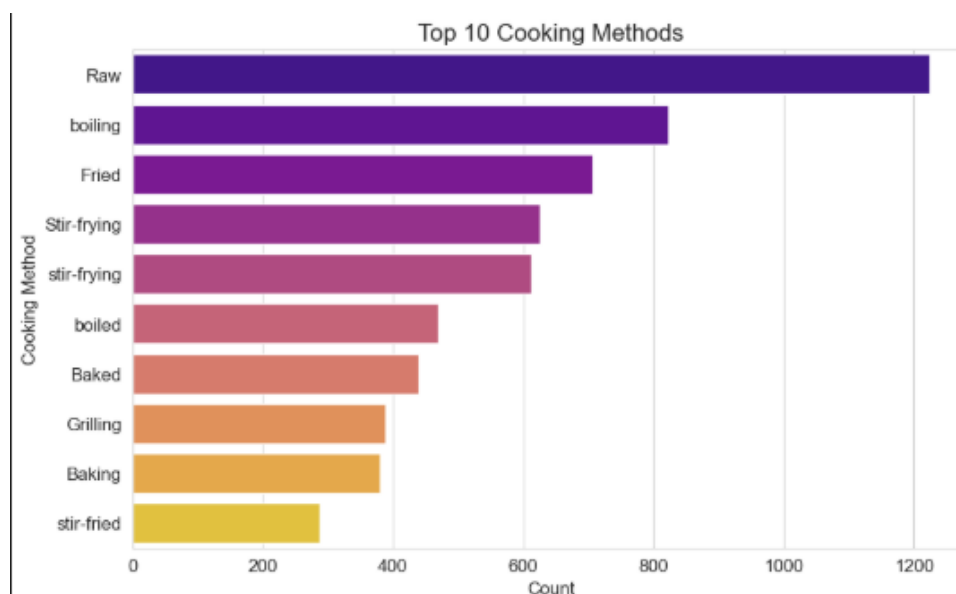


Figure 2: Top 10 Most Common Cooking Methods

4.2 Time Series Analysis

To understand temporal patterns in the data, the number of food submissions over time was plotted. As seen in Figure 3, the data shows a clear and significant peak in submissions during mid-2025, with a sharp drop-off afterward. This suggests that the data might have been collected as part of a specific campaign or event, rather than being a continuous stream.



Figure 3: Number of Food Submissions Over Time

4.3 Nutrient Distribution Analysis

Understanding the distribution of key nutritional values is essential. Figure 4 provides a combined view of the distributions for calories and the three core macronutrients.

- **Calories Distribution:** Both the histogram and box plot show a heavily right-skewed distribution. Most dishes fall below 1000 kcal, but there is a long tail of high-calorie outliers.

- **Macronutrient Distributions:** Protein, fat, and carbohydrates all exhibit similar right-skewed distributions, indicating that most dishes have a moderate amount of these nutrients, with fewer instances of extremely high values.

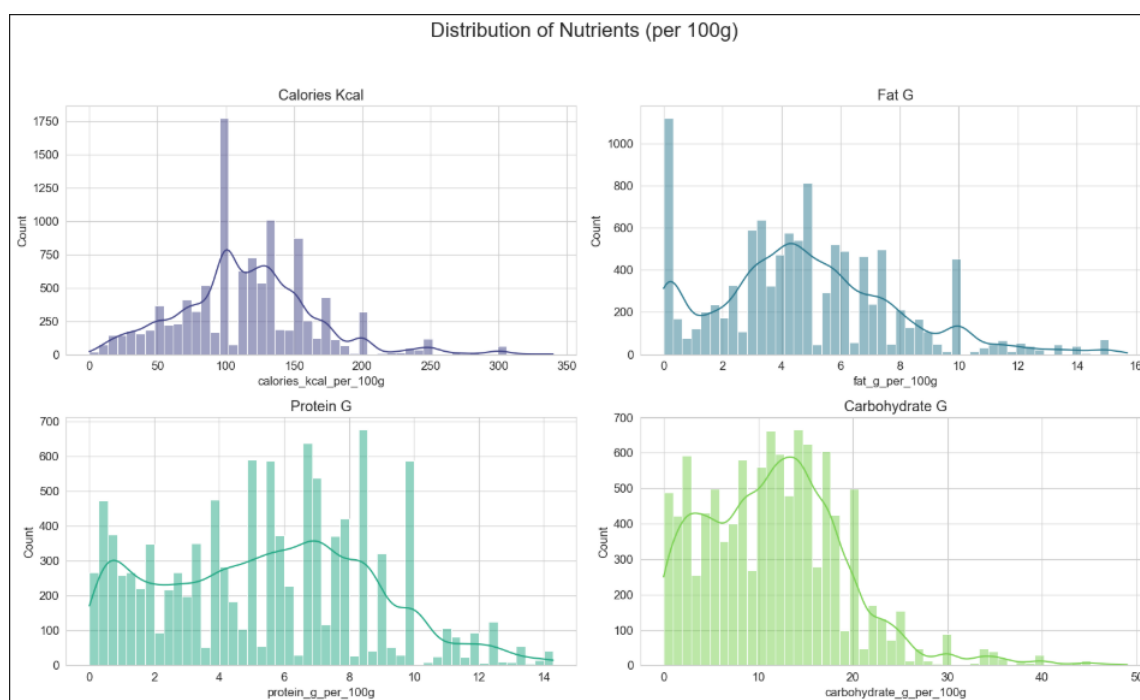


Figure 4: Distribution of Nutrients (Calories, Protein, Fat, Carbs)

To explore the relationships between these numerical features, a correlation matrix was generated (Figure 5). The heatmap visually confirms expected nutritional principles: calories have a strong positive correlation with fat and carbohydrates. `total_grams` also shows a moderate positive correlation with all nutrients, which is intuitive.

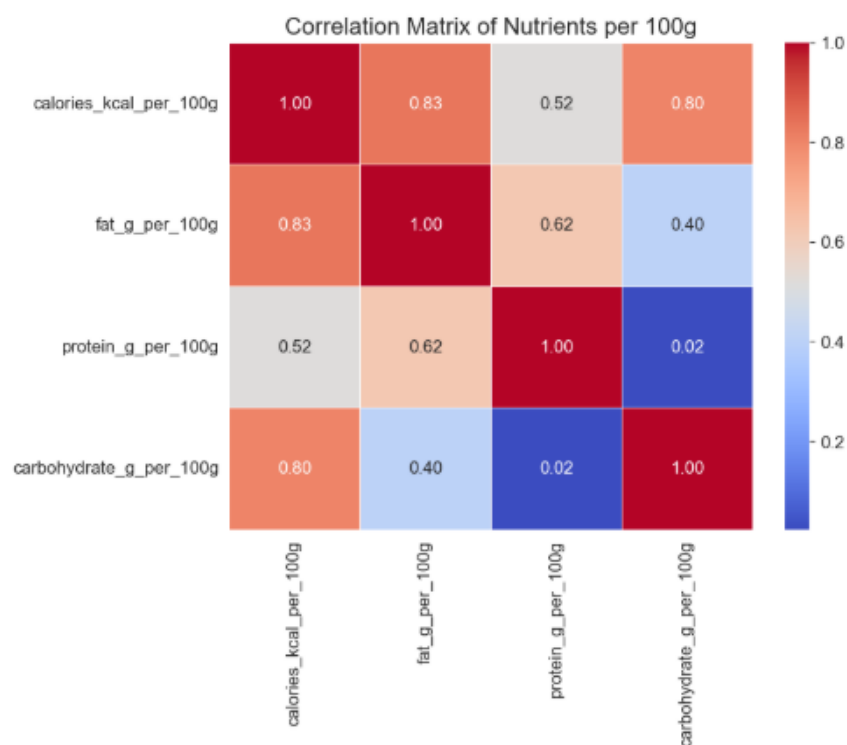


Figure 5: Correlation Matrix of Nutrients

Figure 6 presents box plots for each nutrient, which are useful for comparing their relative spreads and identifying outliers. It confirms that the range of values for calories and carbohydrates is much wider than for protein and fat.

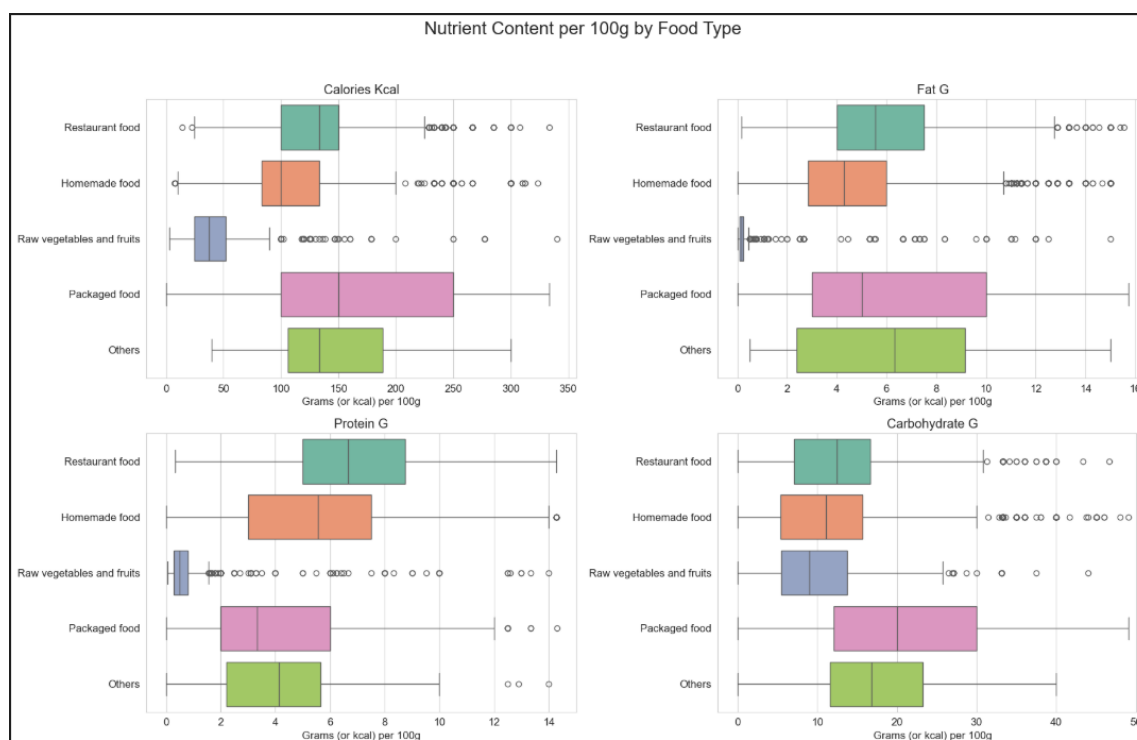


Figure 6: Box Plot of Nutrients

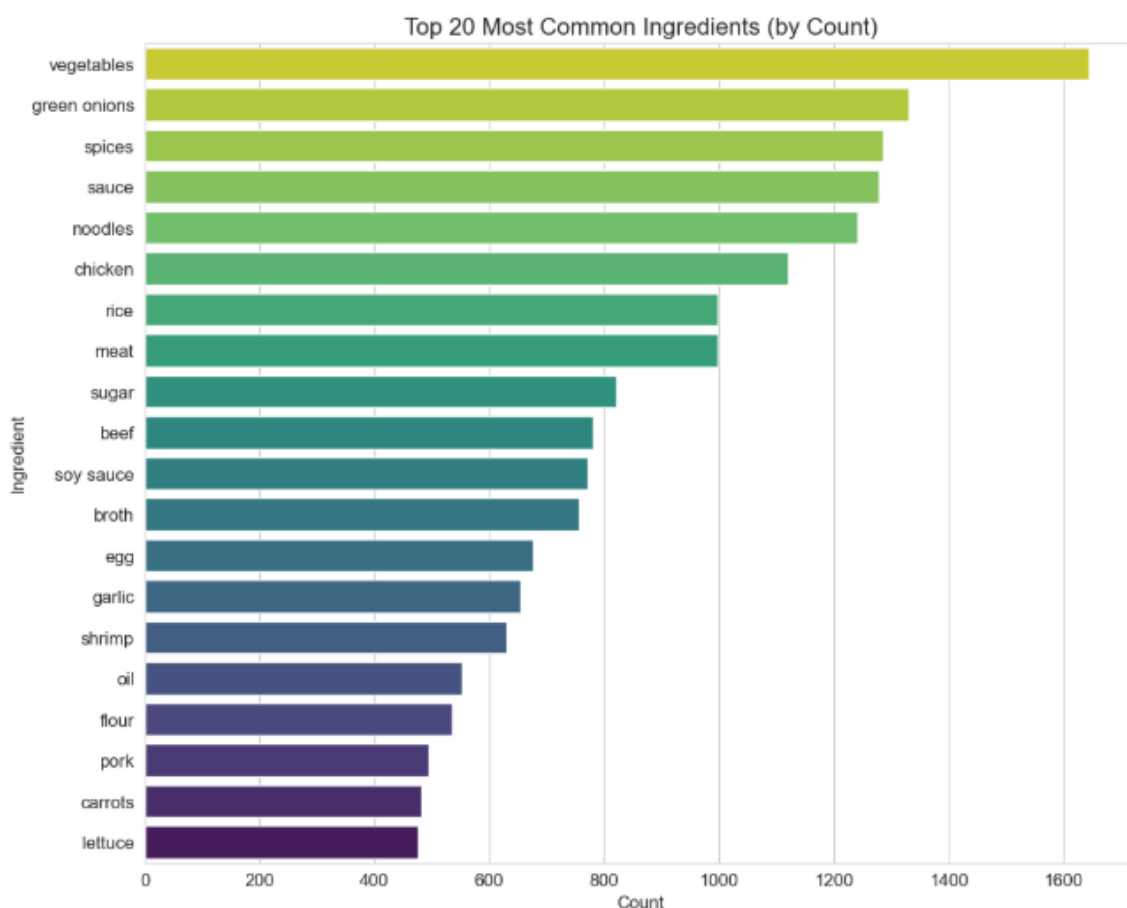


Figure 8: Top 20 Most Common Ingredients

4.5 Image Feature Analysis

Basic properties of the processed images themselves were analyzed. Figure 9 shows the distribution of image dimensions. Most images cluster around the maximum resized dimension of 512 pixels, as expected from the preprocessing step.

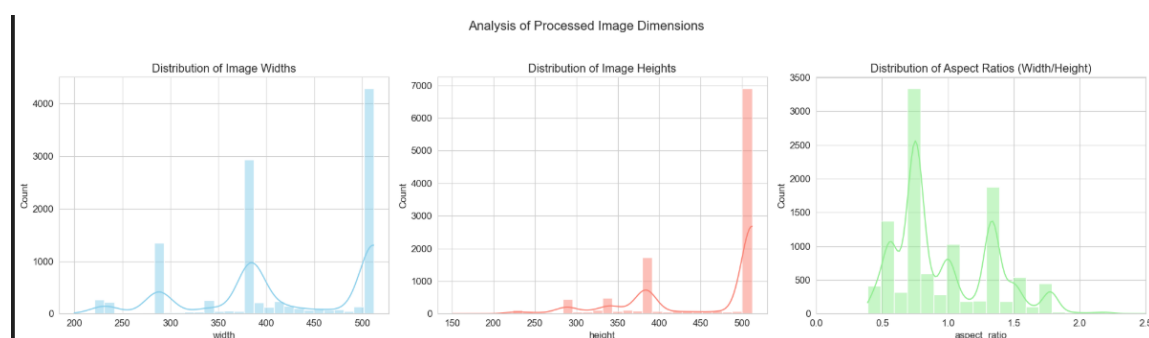


Figure 9: Analysis of Processed Image Dimensions

The distribution of average image brightness is shown in Figure 10. The brightness values appear to follow a roughly normal distribution centered around a mean of approximately 120, indicating that most images are well-lit and neither excessively dark nor bright.

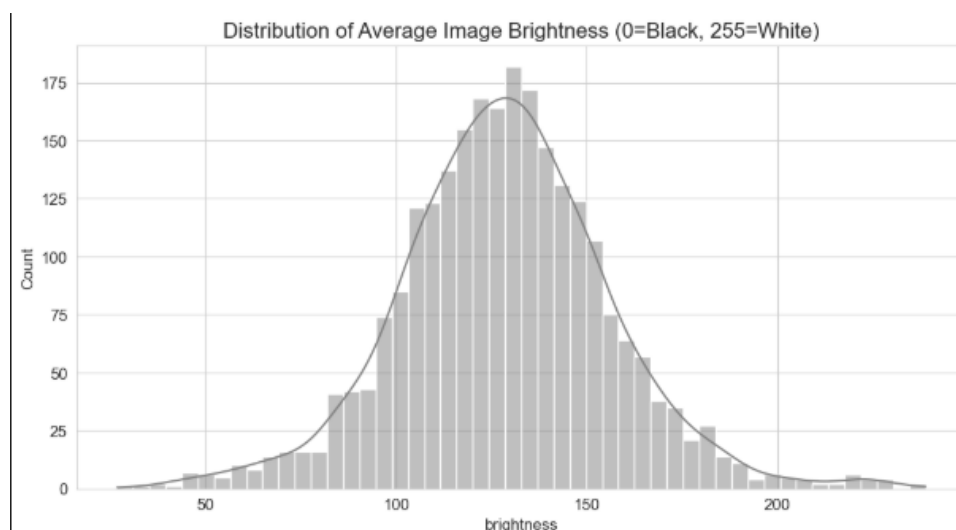


Figure 10: Distribution of Average Image Brightness

Figure 11 displays the average color distribution across all images. It shows that the green and blue channels have very similar distributions, while the red channel has a slightly higher mean value. This suggests a tendency towards warmer tones in the food photography.

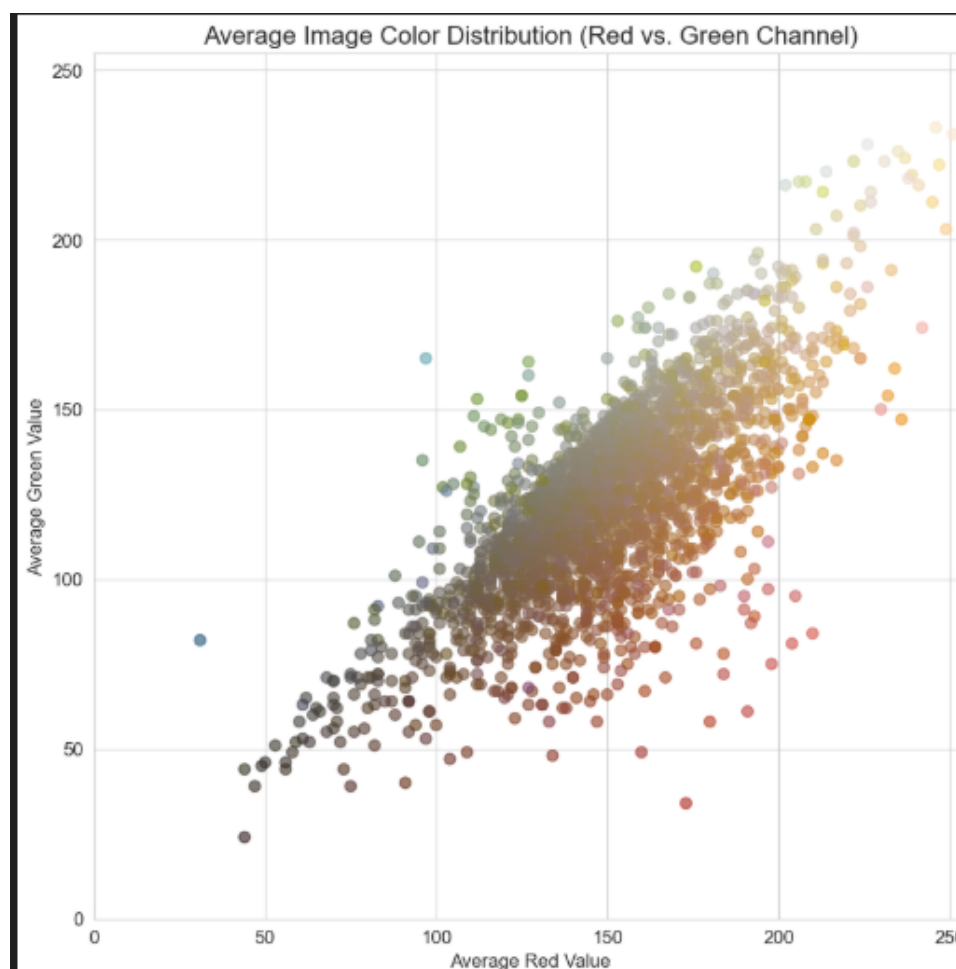


Figure 11: Average Image Colour Distribution

4.6 Qualitative Image Review

Finally, a qualitative review was performed by displaying samples of the images. Figure 12 shows a random sample, confirming that the dataset contains a wide variety of recognizable food images.

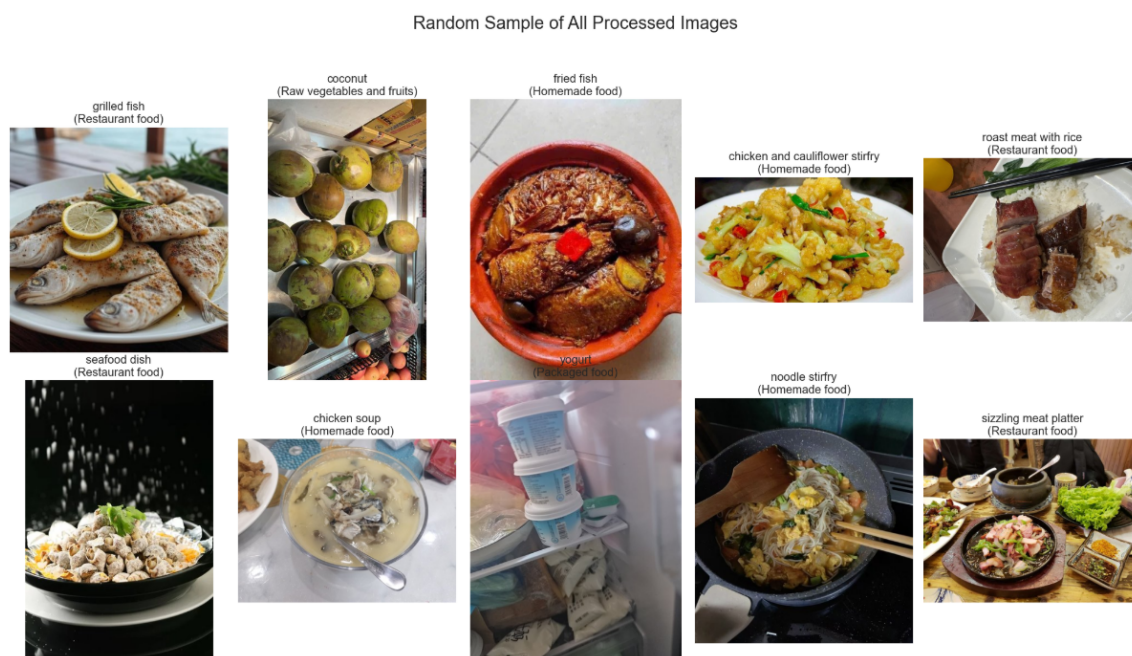


Figure 12: Random Sample of All Processed Images

Figure 13 displays a sample of the darkest images from the dataset, which helps in identifying potential quality issues or specific types of food photography (e.g., meals in dark restaurant settings).

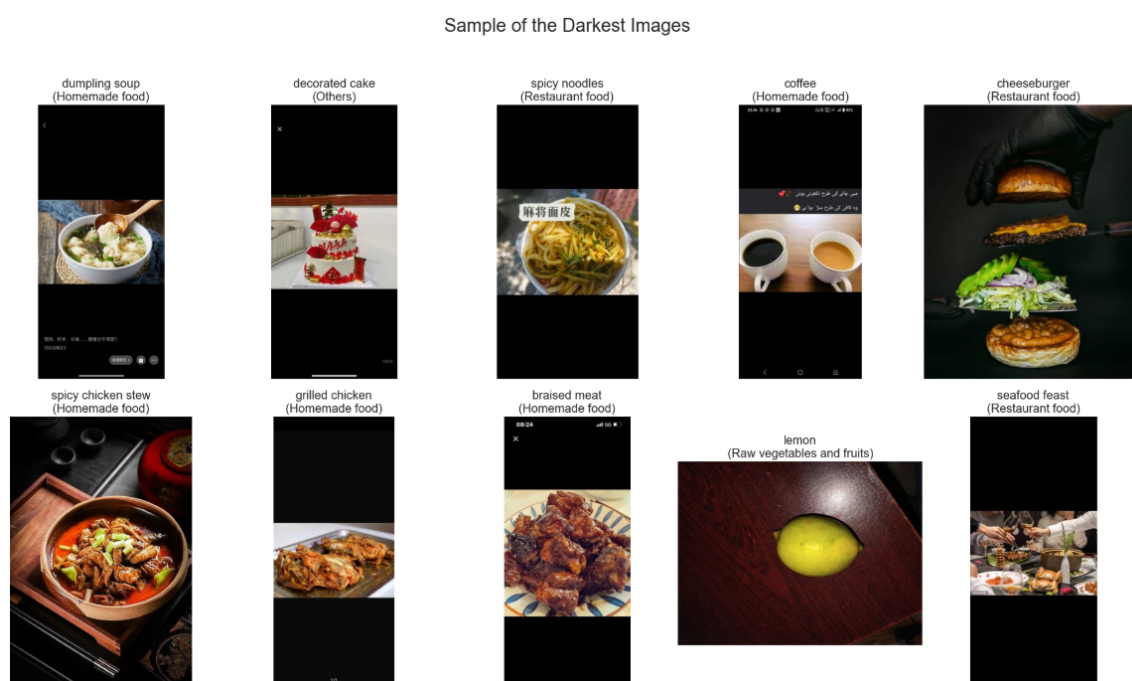


Figure 13: Sample of Darkest Images

Lastly, Figure 14 shows a sample of images corresponding to high-protein foods. This thematic sampling confirms that the textual data aligns well with the visual content, as the images clearly depict items like grilled meats and fish.



Figure 14: Sample of High Protein Foods

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Appendix

Project Links

- **Preprocessing Notebook:** <https://drive.google.com/file/d/1pW9QuCXKXB8r8gG262batrwAPJl4sW4/view?usp=sharing>
- **EDA Notebook:** https://drive.google.com/file/d/1jq3XJvtocd1TvsD_hwkq7xQR3eUYZdWB/view?usp=sharing
- **Dataset Link:** <https://docs.google.com/spreadsheets/d/141AMjFA9DilZQbQSYMDQ7xprvEu601ePzQNou4cbbcg/edit?usp=sharing>
- **GitHub Repository:** <https://github.com/Aaditya235-design/NutriSage>

GitHub Commits

A snapshot of the project's version control history done till now is shown in Figure 15.



Figure 15: Record of GitHub Commits